Titanic - Machine Learning

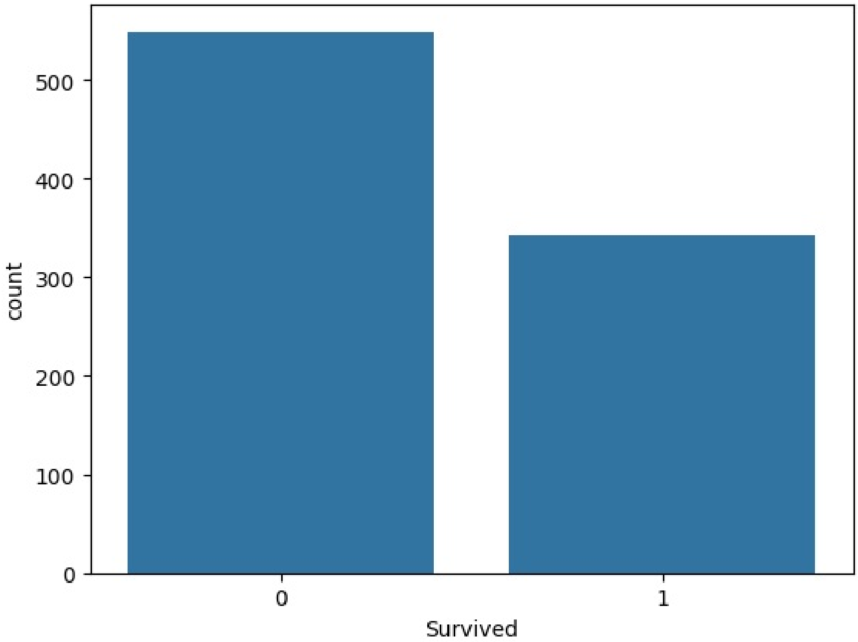
In [136…

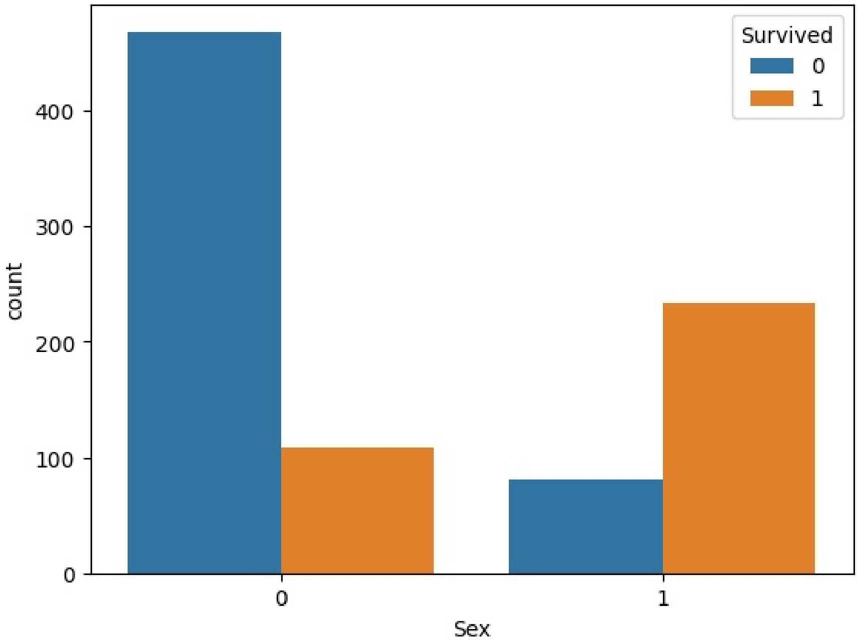
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.metrics import accuracy_score, confusion_matrix

# 1. Load the dataset
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

print(" Data loaded successfully")
print("Dataset size:", df.shape)
print("\nFirst 5 rows:")
print(df.head())

# 2. Basic info about the dataset
print("\n\n Dataset information:")
print(df.info())

print("\n\n Missing values count:")
print(df.isnull().sum())

# 3. Important columns
print("\n\nImportant columns:")
print(df[['Survived', 'Pclass', 'Sex', 'Age', 'Fare', 'Embarked']].head())

# 4. Target variable (Survived)
print("\n\nSurvival counts:")
print(df['Survived'].value_counts())

# 5. Basic preprocessing
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df['Embarked'] = df['Embarked'].fillna('S')
df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})

print("\n Basic preprocessing done")

# 6. Split features and target
features = ['Pclass', 'Sex', 'Age', 'Fare', 'Embarked']
X = df[features]
y = df['Survived']

print("\n Features and target separated")
print("X shape:", X.shape)
print("y shape:", y.shape)

# 7. Train-test split
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("\n Train and test sets created")
print("Train size:", X_train.shape[0])
print("Test size:", X_test.shape[0])

# 8. Build and train the model
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

print("\n Model trained")

# 9. Predictions and accuracy
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print("\n Test Accuracy:", round(accuracy*100, 2), "%")

# Confusion Matrix
print("\n Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Visualization
sns.countplot(x='Survived', data=df)
plt.show()
```

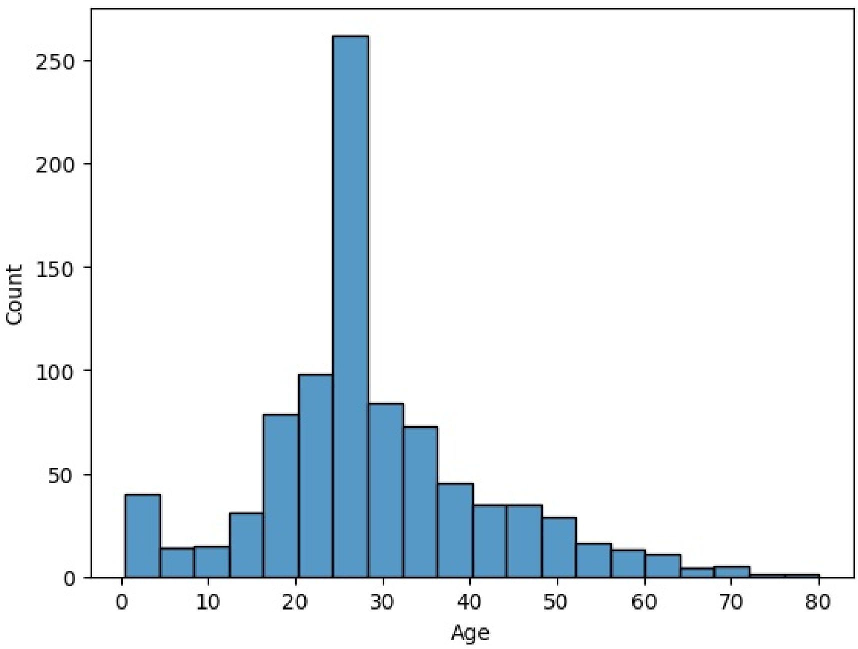
```python
sns.countplot(x='Sex', hue='Survived', data=df)
plt.show()

sns.histplot(df['Age'], bins=20)
plt.show()
```

```
 Data loaded successfully
Dataset size: (891, 12)

First 5 rows:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S


 Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None


 Missing values count:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64


Important columns:
   Survived  Pclass     Sex   Age     Fare Embarked
0         0       3    male  22.0   7.2500        S
1         1       1  female  38.0  71.2833        C
2         1       3  female  26.0   7.9250        S
3         1       1  female  35.0  53.1000        S
4         0       3    male  35.0   8.0500        S


Survival counts:
Survived
0    549
```

```
1    342
Name: count, dtype: int64

 Basic preprocessing done

 Features and target separated
X shape: (891, 5)
y shape: (891,)

 Train and test sets created
Train size: 712
Test size: 179

 Model trained

 Test Accuracy: 79.33 %

 Confusion Matrix:
[[88 17]
 [20 54]]
```







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